Does the Q410/Q417 refutation land on Korff's object?

Scope, domain, and the ribbon escape route

Clio

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`proofs` repo at commit `81e8a49`

Abstract

Three questions about the *scope* of my Q410/Q417 theorems, raised independently within twenty-four hours by my own registry, by the 10–11 dream, and by Rick's referee report (email UID 796). All three are answered here. **(A)** Rick's replacement witness is verified by a mechanism disjoint from his: at $k \geq 2$, $\ell \geq 1$, $n = 2$, $\alpha = (1, 1)$ the Q410 obstruction fails *inside* Korff's stated domain, so the theorems are about his function and not only about his formula. **(B)** Korff's Definition 5.5 carries *no* parameter besides t; the winding is graded by an independent formal variable z sitting *outside* the weight, and his lattice q satisfies $t = q^2$ — it is t, not Macdonald's q. So $q = 0$ is not a substitution that destroys the term doing the work, and Q424 closes NO. **(C)** The ribbon escape route of Korff `1906.02565` survives its pre-registered kill — its weight is not an index-set product and passes the $t = 1$ screen with $W_T(1) = 1$ — and then dies by a *new* mechanism: it assigns the value 1 to every single-box move, hence is t-blind on the fibre $\alpha = (1^n)$, while $c_{(1^4)}(t) = 3 - t^2$ at a point inside Korff's domain. The two failures sit at opposite ends of one invariant, and together they sharpen the specification.

§0 Holdings row, measured before any mathematics

object	path	state
Korff `1110.6356`	`data/arxiv-rag/cylindric-partitions/korff_1110.6356.pdf`	on disk, 1 12
Korff `1906.02565`	`projects/sources/korff-cylindric-20260923/1906.02565-src/draft_02.tex`	on disk, 185
Rick's referee report	`mail/attachments/796/...q410-korff-weight.pdf`	read in full
Rick's scripts	`grandpa-rick/work-in-progress/scripts/day235/q410/`	**not fetche**

Rick's scripts were deliberately *not* fetched. The brief asks for two *mechanisms*, not two handles on one; reading his code first would have made my engine a second reading of his. My two engines are described in §1.2.

1 Task A. Rick's domain-legal witness, verified and adopted

1.1 Setting and dictionary

Fix $k \geq 1$ and $\ell \geq 0$ and set

$$h := 2k, \qquad w := 2\ell + 2, \qquad M := k + \ell + 1, \qquad N := 2M = 2h + w.$$

Under the dictionary of my node `korff-hkko-dictionary` (proved two independent ways), Korff's parameters are $n_K = h$ (cyclic slots) and $k_K = w$ (level). A state is $x \in R := \{x \in \mathbb{Z}^h : x_1 \geq x_2 \geq \cdots \geq x_h \geq x_1 - w\}$, a step is $\theta \in \{0,1\}^h$ with $x \mapsto x + \theta$ staying in R, and Korff's particle configuration is $m_i(x) = x_i - x_{i+1}$ for $i < h$, $m_h(x) = w - (x_1 - x_h)$. A *cylindric tableau* T of content $\alpha \in \mathbb{Z}^n_{\geq 0}$ is a path $x^{(0)} = 0, x^{(1)}, \ldots, x^{(n)}$ in R whose a-th step $\theta^{(a)}$ has α_a ones (HKKO `2301.13117`, PROP:CSSYT=PATH). Write

$$J_{\mathrm{cyc}}(\theta) = \{\, i \in \mathbb{Z}_h : \theta_i = 0,\ \theta_{i+1} = 1 \,\}, \qquad \nu(T) = \sum_a \big|J_{\mathrm{cyc}}(\theta^{(a)})\big|,$$

and let $\Psi_T(t) = \prod_a \prod_{i \in J_{\mathrm{cyc}}(\theta^{(a)})} \big(1 - t^{m_i(x^{(a-1)})}\big)$ be Korff's weight (5.13). Let $c_\alpha(t)$ be the coefficient of x^α in $(x_1 \cdots x_n)^k \,\mathrm{Eq}_{\mathrm{gen}}$ (Warnaar `2511.17034`), and let $c^-_{2k,w}(\lambda) \in \{0, \pm 1\}$ be the HKKO sign of the endpoint.

The theorem being scope-tested is my node `q410-no-cyclic-psi-weight`: *if π_T is cyclically ψ-graded, i.e.* $\mathrm{ord}_{t=1}\, \pi_T \geq \nu(T)$, *then* $\mathrm{ord}_{t=1}\, c_\alpha \geq \min_T \nu(T)$ *over the fibre.* Rick's defect D3 is that my witness family ($k = 1$, $\alpha = (1^{2M})$) violates both of Korff's standing restrictions: he assumes $n_K > 2$, i.e. $k \geq 2$, and his Definition 5.5 sets $P \equiv 0$ for $\ell_K > k_K$, i.e. for $n > w$. My own node `korff-own-range-excludes-the-target` had recorded both, by name, and graded itself RECORDED, NOT USED. See §4.

1.2 Theorem, and a by-hand proof at the critical point

Theorem 1 (domain-legal witness)**.** *Let $k \geq 1$, $\ell \geq 0$, $n = 2$, $\alpha = (1,1)$. Then*

(i) the fibre consists of exactly two cylindric tableaux, with endpoints $\lambda = (2, 0^{h-1})$ and $\lambda = (1,1,0^{h-2})$;

(ii) $\nu(T) = 2$ for both, so $\min_T \nu(T) = 2$;

(iii) $c^-_{2k,w}(1,1,0^{h-2}) = +1$ and $c^-_{2k,w}(2,0^{h-1}) = -[\ell = 0]$, whence $c_\alpha(1) = 1 - [\ell = 0]$;

(iv) consequently, for every $\ell \geq 1$, $\mathrm{ord}_{t=1}\, c_\alpha = 0 < 2 = \min_T \nu(T)$, so the conclusion of `q410-no-cyclic-psi-weight` *fails.*

Moreover $n = 2 \leq w$ always, and $n_K = h = 2k > 2$ as soon as $k \geq 2$: for $k \geq 2$, $\ell \geq 1$ the failure point lies inside both *of Korff's standing restrictions.*

Proof. (i) From $x^{(0)} = 0$, the step e_i gives $x = e_i$, which lies in R iff $x_{i-1} \geq x_i$, i.e. iff $i = 1$ (for $i \geq 2$ one would need $0 \geq 1$); the cylindric condition $x_h \geq x_1 - w$ reads $0 \geq 1 - w$, true as $w \geq 2$. So $\theta^{(1)} = e_1$. From $x = e_1$, the step e_1 gives $(2, 0^{h-1})$: weak decrease holds and $0 \geq 2 - w$ holds since $w \geq 2$. The step e_2 gives $(1,1,0^{h-2})$: weak decrease holds and $0 \geq 1 - w$ holds (for $h = 2$ read $(1,1)$, and $x_2 = 1 \geq 1 - w$). For $i \geq 3$ one would need $x_2 = 0 \geq x_i = 1$. So exactly two tableaux.

(ii) Each step is a unit vector e_i in $h \geq 2$ cyclic slots, hence a non-constant cyclic 0/1 word; $J_{\mathrm{cyc}}(e_i) = \{i - 1 \bmod h\}$ has exactly one element. Two steps give $\nu(T) = 2$.

(iii) HKKO's sign is $+1$ iff $\lambda_{2i-1} = \lambda_{2i}$ for $1 \leq i \leq k$, and -1 iff $\lambda_1 - \lambda_{2k} = w$ and $\lambda_{2i} = \lambda_{2i+1}$ for $1 \leq i \leq k-1$. For $\lambda = (1,1,0^{h-2})$ the pairs are $(1,1), (0,0), \ldots, (0,0)$, so the sign is $+1$. For $\lambda = (2, 0^{h-1})$ the first pair is $(2,0)$, so the $+1$ clause fails; the -1 clause needs $\lambda_1 - \lambda_h = 2 = w$, i.e. $\ell = 0$, and $\lambda_{2i} = \lambda_{2i+1} = 0$, which holds. Hence the sign is $-[\ell = 0]$. By HKKO Theorem 3.3, $c_\alpha(1) = \sum_\lambda c^-(\lambda)\, K^\alpha_{\lambda[2k,w]'}$ and each endpoint is reached by exactly one path, so $c_\alpha(1) = 1 - [\ell = 0]$.

(iv) For $\ell \geq 1$, $c_\alpha(1) = 1 \neq 0$, so $\mathrm{ord}_{t=1}\, c_\alpha = 0$, and $0 < 2$. □

Remark 2. Part (iv) is complete *by hand* for $\ell \geq 1$ and needs no computation: a nonzero value at $t = 1$ already pins the order to 0. For $\ell = 0$ one has $c_\alpha(1) = 0$ and the order must be read off the polynomial; §1.3 measures $c_\alpha(t) = 1 - t$, order 1, still < 2. I record the division of labour because it is the honest one: the $\ell \geq 1$ half of Theorem 1 is a proof, the $\ell = 0$ half rests on computation.

1.3 Verification: two mechanisms, not two handles

M1 (symmetric functions). $c_\alpha(t)$ as the coefficient of x^α in $(x_1 \cdots x_n)^k\, G_t$, with $G_t = \sum_{y \in \mathbb{Z}^k} \det_{1 \leq i,j \leq k}\big(t^{My_i^2 - \dots}$ $\dot{e}_{n+i+j-Ny_i}\big)\big)$, $\dot{e}_r = e_r(x_1, x_1^{-1}, \dots, x_n, x_n^{-1})$. Exact rational arithmetic in SymPy. This engine never constructs a lattice path.

M2 (signed lattice paths). $A - B$ where A (resp. B) counts HKKO paths of content α whose endpoint has $c^- = +1$ (resp. -1). This engine never constructs a determinant.

The two share no code path. Results at $n = 2$, $\alpha = (1, 1)$:

k	ℓ	h	w	$\#T$	$\min \nu$	$c_\alpha(t)$ [M1]	$\mathrm{ord}_{t=1}$	$c_\alpha(1)$ [M2]	in Korff's domain
1	0	2	2	2	2	$1 - t$	1	0	no ($n_K = 2$)
1	1, 2, 3	2	4, 6, 8	2	2	1	0	1	no ($n_K = 2$)
2	0	4	2	2	2	$1 - t$	1	0	**yes**
2	1, 2, 3	4	4, 6, 8	2	2	1	0	1	**yes**
3	0	6	2	2	2	$1 - t$	1	0	**yes**
3	1, 2, 3	6	4, 6, 8	2	2	1	0	1	**yes**
4	0	8	2	2	2	$1 - t$	1	0	**yes**
4	1, 2, 3	8	4, 6, 8	2	2	1	0	1	**yes**

Measured totals. M1 and M2 agree at $t = 1$ in **16 of 16** pairs. $c_{(1,1)} \in \{1 - t, 1\}$ in **16 of 16**, exactly the set Rick reports from `witness_n2.py` — reproduced here by a disjoint mechanism. The obstruction fires ($\mathrm{ord}_{t=1} c_\alpha < \min_T \nu$) in **12 of 12** domain-legal rows. Korff's identity $\sum_T c^-(T)\Psi_T(t) = c_\alpha(t)$ fails in **16 of 16**.

At the single critical point $k = 2$, $\ell = 1$ — the smallest point that is simultaneously domain-legal ($n_K = 4 > 2$) and has $n = 2 \leq w = 4$ — the whole fibre is:

$$\begin{aligned} \lambda = (2,0,0,0): &\quad c^- = 0, \quad \nu = 2, \quad \Psi_T = (t-1)^2(t+1)(t^2+1)(t^2+t+1),\ \Psi_T(1) = 0, \\ \lambda = (1,1,0,0): &\quad c^- = +1, \quad \nu = 2, \quad \Psi_T = (t-1)^2(t+1)(t^2+1),\ \Psi_T(1) = 0, \end{aligned}$$

so $\sum_T c^-(T)\Psi_T(t) = (t-1)^2(t+1)(t^2+1)$, which is 0 at $t = 1$, whereas $c_\alpha(t) = 1$. *The refutation is a single evaluation at a single point inside Korff's domain.*

1.4 The three pre-registered controls, each reported as a number

(i) $\nu = 0$ must make the obstruction vanish. $\nu(T) = 0$ requires every $\theta^{(a)}$ to be constant, i.e. $\alpha_a \in \{0, h\}$. Four arms:

arm	$\min \nu$	$c_\alpha(t)$	obstruction vacuous	Korff's identity holds
$k = 1, \ell = 0, \alpha = (2),\ h = 2$	0	1	**yes**	**yes**
$k = 1, \ell = 1, \alpha = (2),\ h = 2$	0	1	**yes**	**yes**
$k = 2, \ell = 1, \alpha = (4),\ h = 4$	0	1	**yes**	**yes**
$k = 2, \ell = 1, \alpha = (2),\ h = 4$	1	1	no	**no**

This control *fires*, and in the strongest available sense: wherever $\min \nu = 0$ Korff's identity holds *exactly*, and the one arm whose step is non-constant ($\alpha = (2)$ at $h = 4$, so $2 \notin \{0, 4\}$) fails. The obstruction is therefore located, not blanket. Note the brief's description of this control was slightly off: $\alpha = (2)$ is a constant step only when $h = 2$; at $k = 2$ the constant arm is $\alpha = (4)$.

(ii) $\ell = 0$ and $\ell \geq 1$ must differ. At $k = 1, 2, 3, 4$: $\ell = 0$ gives $c = 1 - t$ with $\mathrm{ord}_{t=1} = 1$; $\ell = 1$ gives $c = 1$ with $\mathrm{ord}_{t=1} = 0$. Orders distinct in **4 of 4**. The $[\ell = 0]$ sign term is being applied.

(iii) Regression on the old, out-of-domain family. $k = 1$, $\alpha = (1^{2M})$: M2 gives $c_\alpha(1) = 0, 1, 8$ at $M = 2, 3, 4$, matching the Q405 closed form $C_M - (2M - 1)t^{M-1} + t^{M+1}$ evaluated at $t = 1$ in **3 of 3**. This arm is a regression check, *not* evidence — $n = 2M > w$ there, so Korff sets $P \equiv 0$.

Corollary 3. *Rick's repair holds. Pre-registered outcome* "repair holds" *is realised, with the sharpening that it holds for* all $k \geq 1$ *(not only $k \geq 2$) as a statement about the formula, and for $k \geq 2$ as a statement inside Korff's domain. Q410 and Q417 survive with a different witness.*

2 Task B. Q424: which parameter grades the wrap in Definition 5.5?

This was a *read*, of a PDF that had been on my disk for months. Korff `1110.6356`, §5.3. The three questions, answered.

(a) Does Definition 5.5's weight contain a parameter other than t? No. Equations (5.12) and (5.13) read, verbatim,

$$\Phi_{\lambda/d/\mu}(t) = \begin{cases} \prod_{i \in I_{\lambda/d/\mu}} (1 - t^{m_i(\lambda)}), & \lambda/d/\mu \text{ a horizontal strip} \\ 0 & \text{else,} \end{cases} \qquad \Psi_{\lambda/d/\mu}(t) = \begin{cases} \prod_{i \in J_{\lambda/d/\mu}} (1 - t^{m_i(\mu)}), & \text{ditto} \\ 0 & \text{else,} \end{cases}$$

with $i \in I$ iff $\lambda'[d]_i - \mu'[0]_i = 1$ and $\lambda'[d]_{i+1} - \mu'[0]_{i+1} = 0$, and $i \in J$ iff $\lambda'[d]_i - \mu'[0]_i = 0$ and $\lambda'[d]_{i+1} - \mu'[0]_{i+1} = 1$. Definition 5.5 then sets $Q_{\lambda/d/\mu}(x_1, \ldots, x_\ell; t) = \sum_{|T| = \lambda/d/\mu} \Phi_T(t) x^T$ and likewise for P with Ψ_T. Only t and the x_i occur. The wrap enters the *index set* — the index i runs over $1 \leq i \leq n$ with $i + 1$ read cyclically, Korff's own proof of Lemma 5.4 noting "$\theta_{n+1} = \lambda'[d]_{n+1} - \mu'[0]_{n+1} = \lambda'[d]_1 - \mu'[0]_1$" — and through m_n, which is a power of t. There is nothing else to specialise.

But the ambient object does carry one, and it is z, not q. Immediately before Lemma 5.1, Korff writes: *"We introduce a formal variable z keeping track of the winding number around the cylinder; the latter is the same variable as the one used in (3.39) and (3.42)"*, where $E(u) = A(u) + z\, D(u) = \sum_r u^r e_r$ is (3.42). Lemma 5.5, equation (5.17), is then

$$e_r |\mu\rangle = \sum_{\lambda/d/\mu = (r)} z^d \, \Phi_{\lambda/d/\mu}(t) \, |\lambda\rangle = \sum_{\mu/d/\lambda = (n-r)} z^{1-d} \, \Psi_{\mu/d/\lambda}(t) \, |\lambda\rangle, \qquad d \in \{0, 1\}.$$

So the winding *is* graded, by z^d, and z *is* free: in the proof of Lemma 5.1 Korff states *"since z is an independent formal variable (the weights of the lattice configurations do not depend on z)"*. That parenthesis is the answer to Q424 in one clause: the grading of the wrap lives *outside* the weight, as a monomial prefactor, and Korff sets $z = 1$ from §7 onwards (Theorems 7.8, 7.11, 7.13, 7.14, 7.16 all open "set $z = 1$").

(b) Is it free where Warnaar's question has content? Free — and irrelevant.

Proposition 4 (no regular prefactor repairs the identity)**.** *Let $\mathcal{F}$ be a fibre of cylindric tableaux with $\Psi_T(1) = 0$ for every $T \in \mathcal{F}$ and $c_\alpha(1) \neq 0$. Then for any family of functions u_T regular at $t = 1$, $\sum_{T\in\mathcal{F}} c^-(T)\, u_T\, \Psi_T(t) \neq c_\alpha(t)$.*

Proof. Evaluate at $t = 1$: the left side is $\sum_T c^-(T) u_T(1) \cdot 0 = 0$, the right side is $c_\alpha(1) \neq 0$. □

The fibre of Theorem 1 at $k = 2, \ell = 1$ satisfies the hypothesis ($\Psi_T(1) = 0$ for both tableaux, $c_\alpha(1) = 1$). Taking $u_T = z^{d(T)}$ gives: the z-graded refinement of Definition 5.5 fails for *every* value of z, free or specialised, and in particular Korff–Vasilev's $z = \pm q = \pm\sqrt{t}$ (regular at $t = 1$, value ± 1) does not help. The only escape Proposition 4 leaves open is a prefactor with a *pole* at $t = 1$ of order $\geq \nu(T)$ — that is, a z which is not a formal variable at all but a t-dependent substitution singular at $t = 1$. No such object occurs in either paper.

(c) Is Korff's q Macdonald's q? No, and this is where "specialised" lives. Two distinct symbols are both written q in the literature around this paper.

- **Macdonald's** q is the one the title specialises. The abstract: *"We define cylindric generalisations of skew Macdonald functions $P_{\lambda/\mu}(q, t)$ when either $q = 0$ or $t = 0$."* At $q = 0$ one gets Hall–Littlewood $P_\lambda(x; t)$, and §1 confirms the identification explicitly, in the Satake map $P_\lambda(Y_1, \ldots, Y_r; 0, t = q^2)$ with the gloss "where $P_\lambda(0, t)$ are the Hall–Littlewood polynomials". So Definition 5.5 is the $q_{\mathrm{Mac}} = 0$ branch, and the Hall–Littlewood t is what survives.
- **Korff's lattice/Hecke** q is *not* independent of t. §2.1: *"q plays here the role of a dummy variable and we will apply the same definitions for other indeterminates or powers of q; in particular we will often use $t = q^2$ instead of q."* Theorem 1.6 is stated "For $t = q^2$". So this q is $\sqrt{t}$: the same deformation parameter, renamed.

Hence the "two specialisations" of the abstract are the two *branches* $q_{\mathrm{Mac}} = 0$ and $t = 0$ of Macdonald's two-parameter family — not two nested specialisations of one weight. Definition 5.5 has exactly one deformation parameter.

Verdict. Q424 closes no. My Q410/Q417 theorems are about Korff's object as repaired by Task A, not about a specialisation of it. Setting $q_{\mathrm{Mac}} = 0$ is what *creates* the Hall–Littlewood weight I refuted, not a collapse of a term that was doing work; and the one genuinely free parameter grading the wrap, z, is excluded by Proposition 4.

Remark 5. This is the outcome the brief told me to want least, because it is the one I would reach for if I stopped reading early. So I record what the extra reading actually bought: stopping at (5.13) would have given "no parameter besides t" and *missed z entirely.* The honest answer is not "there is no second parameter" — there is one, it is named, it is free, and it grades exactly the wrap — but "the second parameter is a monomial prefactor and Proposition 4 kills it". Those are different sentences, and only the second is true.

3 Task C. Q418: the ribbon route survives its kill and dies of something else

3.1 The pre-registered kill does not fire

The kill was: *if `1906.02565`'s weight is again a product of* $(1-t^m)$ *over a cyclic index set, Q418 dies in twenty minutes by Q410's own theorem.* It is not. Read from `draft_02.tex` (line 466, equation HECKE_WT), for a broken rim hook b with connected components $h \in b$:

$$\mathrm{wt}(b) = (t-1)^{\#b-1} \prod_{h \in b} (-1)^{r(h)-1} t^{c(h)-1},$$

with $r(h)$, $c(h)$ the number of rows and columns h occupies and $\#b$ the number of components; the cylindric extension (line 1258) is the same formula with $\#b$ counting *distinct* cylindric rim hooks and r, c counted in rows $1 \le i \le k$ of $\lambda[d]$. This is not an index-set product: $(t-1)^{\#b-1}$ is a *connectivity* factor, and $t^{c(h)-1}$ a monomial, where Korff's Ψ had a *fold* factor $(1-t^{m_i})$ per cyclic ascent.

Specialising to a horizontal strip — every component is a single row, so $r(h) = 1$ — gives

$$\mathrm{wt}(\text{strip with } p \text{ components of column-widths } c_1, \ldots, c_p) = (t-1)^{p-1} \prod_j t^{c_j - 1}.$$

p	widths	wt	wt(1)
1	(1)	1	1
1	(2)	t	1
1	(3)	t^2	1
2	(1, 1)	$t-1$	0
2	(2, 1)	$t(t-1)$	0
3	(1, 1, 1)	$(t-1)^2$	0

A single box is one rim hook with $|h| = 1$, $r(h) = c(h) = 1$, $\#b = 1$, so

$$\boxed{\mathrm{wt}(\text{single box}) = (t-1)^0 (-1)^0 t^0 = 1 \text{ identically.}}$$

It is a connected move, so the connectivity factor is empty. Hence wt *passes* the screen of `weight-must-be-one-at-t1`, which demanded $W_T(1) = 1$: it attains exactly 1, not merely something nonzero. The capacity argument the brief stated — "an ℓ-ribbon is one connected move, so there is no per-index product and no index set to screen" — is correct, and the five-member index-set family exhausted by `index-set-family-exhausted` does not contain this weight. **Q418 survives its pre-registered kill.**

3.2 And then dies, in Korff's domain, of t-blindness

Theorem 6 (single-box blindness)**.** *Let W be any weight on cylindric tableaux that is multiplicative over steps, $W_T = \prod_a \omega(x^{(a-1)}, \theta^{(a)})$, and suppose that the single-box values $\omega(x, e_i)$ all lie in a ring R not containing t (i.e. are independent of t). Then on any fibre $\alpha = (1^n)$ we have $W_T \in R$ for every T, hence $\sum_T c^-(T) W_T \in R$ is independent of t. Consequently, if $c_{(1^n)}(t)$ is non-constant in t, no such W satisfies $\sum_T c^-(T) W_T = c_{(1^n)}(t)$.*

Proof. Every step of a tableau of content (1^n) adds a single box, so every factor of W_T is one of the $\omega(x, e_i) \in R$ and $W_T \in R$. The signs $c^-(T)$ are integers, so the signed sum lies in R and carries no t. □

Corollary 7 (the ribbon architecture fails in Korff's domain)**.** *At $k = 2$, $\ell = 1$, $n = 4$ — so $h = 4 > 2$ and $n = 4 \le w = 4$, both of Korff's restrictions satisfied — the fibre has* 10 *tableaux and*

$$c_{(1^4)}(t) = 3 - t^2 \quad \textit{(Warnaar engine M1), non-constant,}$$

while Korff's ribbon weight gives $\sum_T c^-(T) W_T = \sum_T c^-(T) = 2$, a constant. The same holds at $k = 3$, $\ell = 1$, $n = 4$.

The q-twist does not rescue it. Korff's cylindric Hecke characters are defined (`1906.02565`, eq. CYLCHI) by $q^d \chi_t^{\lambda/d/\mu}(\nu) = \langle \lambda' | H_{\nu_1} \cdots H_{\nu_\ell} | \mu' \rangle / (t-1)^{\ell(\nu)}$, so the wrap is again graded by a monomial, now in the Novikov parameter q of $qH^*(\mathrm{Gr}_k(\mathbb{C}^n))$, which is t-free. Reading d from Korff `1110.6356`'s proof of Theorem 5.6 (the row configuration is $\sigma = (\sigma^1, \ldots, \sigma^n, \sigma^1)$ with "$\sigma^1 := \sigma^{n+1} := d$" and $\sigma^i = \theta_i$, so $d_a = \theta_1^{(a)}$ and $d(T) = x_1^{(n)}$) gives, at $k = 2, \ell = 1, n = 4$,

$$\sum_T c^-(T)\, q^{d(T)} = -q^4 + 2q^2 + q, \qquad \text{target } 3 - t^2.$$

There is no constant q solving this (the left side is t-free), and $q^4 - 2q^2 - q + 3 - t^2$ is irreducible over $\mathbb{Q}(t)$, so a solution would force q to be a degree-4 algebraic function of t — not the free quantum parameter it is. By Theorem 6 the conclusion is independent of the reading of d: any t-free twist fails.

Verdict. Q418 closes no, by a mechanism the brief did not predict. The pre-registered null ("another index-set product") did not occur; the pre-registered *survival* occurred, the weight passed the screen, and a second, independent obstruction then killed it. Recording this distinction matters: the screen is not the only gate, and a candidate passing it is not thereby alive.

4 The sharpened specification

The two failures are at opposite ends of one invariant. On the fibre $\alpha = (1^4)$ at $k = 2, \ell = 1$, with target $c_\alpha(t) = 3 - t^2$:

weight	single-box value $\omega(x, e_i)$	$\sum_T c^-(T) W_T$	why it fails
required	—	$3 - t^2$	$\mathrm{ord}_{t=1} = 0$ *and* $\deg_t = 2$
Korff 2011, Ψ (5.13)	$(1 - t^{m_i})$, vanishes at $t = 1$	$(t-1)^4(t+1)(t^2+1)(t^2+t+2)$	$\mathrm{ord}_{t=1} = 4$, too **high**
Korff 2019, wt	1, t-free	2	$\mathrm{ord}_{t=1} = 0$ but $\deg_t = 0$, too **lo**

Specification 8. On a fibre $\alpha = (1^n)$ with $c_\alpha(1) \neq 0$ and $c_\alpha(t)$ non-constant, a multiplicative weight $W_T = \prod_a \omega(x^{(a-1)}, \theta^{(a)})$ must have single-box values $\omega(x, e_i)$ that are

(a) *not uniformly vanishing at $t = 1$* — precisely, at least one T in the fibre must have $W_T(1) \neq 0$, since $\sum_T c^-(T) W_T(1) = c_\alpha(1) \neq 0$; and

(b) *not t-free*, by Theorem 6.

Korff 2011 violates (a); Korff 2019 violates (b). A Gaussian binomial $\binom{m+1}{1}_t = (1-t^{m+1})/(1-t) = 1 + t + \cdots + t^m$ satisfies both: its value at $t = 1$ is $m + 1 \neq 0$ and it is non-constant for $m \geq 1$.

This promotes the closing guess of the Q410 paper — a Kostka–Foulkes / modified Hall–Littlewood normalisation, i.e. a change of basis rather than a new statistic — from a hunch to the unique surviving *type* among those examined, by two independent exclusions rather than one. The move from $(1 - t^m)$ to $(1 - t^{m+1})/(1 - t)$ is exactly the division by $(1 - t)$ that turns a fold factor (zero at $t = 1$) into a counting factor (value $m + 1$ at $t = 1$), and it is precisely the correction both of Korff's weights miss, in opposite directions.

5 What the referee found that my own registry had already recorded

This is the finding of the day, and it is about me.

My node `korff-own-range-excludes-the-target` records *both* of Rick's defect-D3 restrictions, by name: "(i) He assumes $n_K > 2$ throughout … the witness family has $k = 1$ and sits at $n_K = 2$. (ii) His Def 5.5 defines $P_{\lambda/d/\mu}(x_1 \dots x_{\ell_K}; t)$ for $1 \leq \ell_K \leq k_K$ and SETS IT TO ZERO for $\ell_K > k_K$ … the witness has $n = N = 2k + w > w$." Its trust grade, in my own hand, is RECORDED, NOT USED.

So Rick did not find something I had missed. He found something I had *written down and declined to let bind*. That is not a coverage failure — the measurement was taken. It is not a consultation failure — the record was read; the node exists because I read it. It is a **decision** failure: I graded an obstruction as observed-and-inert and then built a theorem across it, on the stated ground that "answering a mathematical question by a convention is not an answer". That ground was not wrong. What was wrong was treating it as a reason not to *also* prove the in-domain statement, which Theorem 1 now shows costs two lattice paths and half a page.

The operational repair is not in the record, which was already correct. It is in the reading rule: **a node graded "not used" must be re-read whenever the claim above it is pushed, cited, or refereed.** A grade of "not used" is an unexercised decision, and an unexercised decision is the one place where a correct record and a false paper can coexist.

Two smaller instances of the same shape, recorded for the dream cycle:

- The brief's own description of control (i) was off: $\alpha = (2)$ is a constant step only at $h = 2$. Running it at $k = 2$ as written would have produced a *non*-vacuous arm labelled as the vacuous control, i.e. a control that reads as not firing while measuring the wrong thing. Caught by printing $\min \nu$ as a number rather than trusting the label.

- **The trajectory emitter recorded the opposite of the evidence.** The session's strongest verification — two disjoint mechanisms agreeing in 16/16 pairs, with three controls each reported as a number — was emitted as a `verify` event *without* a `verification_target`. That makes it schema-invalid, so it was **dropped**; and the claim it supports lists it as an input, so that claim reads `tau=unsupported`. Appending four correctly-formed `verify` events afterwards moved **nothing**: τ is a function of a claim's own `inputs` chain at emission time, not of the `verify` events pointing at it. An append-only log cannot be repaired in place. So: *the best-evidenced claim of the session is the one the trajectory grades lowest*, and the cause is

entirely mine. The rule this yields is narrow and checkable — a `verify` must carry its target on the first emission, and a claim must list its verifier as an `input`, not merely be its target.

- Formatting is an instrument. Two scripts died on `TypeError: unsupported format string passed to One.__format__` — SymPy's `Integer(1)` rejects an f-string width spec. Both failures were loud. The dangerous sibling, recorded on 10–08, was silent: `!=` separating $t^3+t^2+t-1.0$ from t^3+t^2+t-1. A third note for the same file: `pkill -f taskA2.py` matched *its own shell's* command line and killed the job it was protecting — exit 144, twice, before I read the cause.

6 Gaps, and what is and is not proved

1. **Proved by hand.** Theorem 1(i)–(iv) for $\ell \geq 1$; Proposition 4; wt(single box) = 1; Theorem 6; Specification 8.

2. **Rests on computation.** Theorem 1 at $\ell = 0$ ($c_\alpha(t) = 1-t$, order 1); Corollary 7's value $c_{(1^4)}(t) = 3-t^2$; the irreducibility of $q^4-2q^2-q+3-t^2$ over $\mathbb{Q}(t)$. Each is a single exact-arithmetic evaluation, cross-checked at $t=1$ by the disjoint mechanism M2.

3. **Gap, named.** Corollary 7 refutes the cylindric broken-rim-hook *weight* transported onto HKKO cylindric tableaux via the single-box value. It does *not* refute Korff `1906.02565`'s construction as a whole, which computes cylindric Hecke characters $\chi_t^{\lambda[d]}$ — a different object, over a different tableau set (broken rim hooks, not horizontal strips). What is proved is: *no weight whose single-box value is t-free can answer Warnaar's question*, and Korff's ribbon weight has a t-free single-box value. Whether some other functional of a cylindric ribbon tableau answers it is open, and is now constrained by Specification 8(b).

4. **Gap, named.** The identification $d_a = \theta_1^{(a)}$ rests on one line of Korff's proof of Theorem 5.6. Theorem 6 is independent of it; only the explicit polynomial $-q^4+2q^2+q$ depends on it.

5. **Not attempted.** Whether a Gaussian-binomial single-box weight actually reproduces $3-t^2$ on this fibre. Specification 8 says it is not excluded; it does not say it works. That is the next session's arithmetic, and the fibre is small enough (10 tableaux, 4 with $c^- \neq 0$) to decide it by linear algebra.